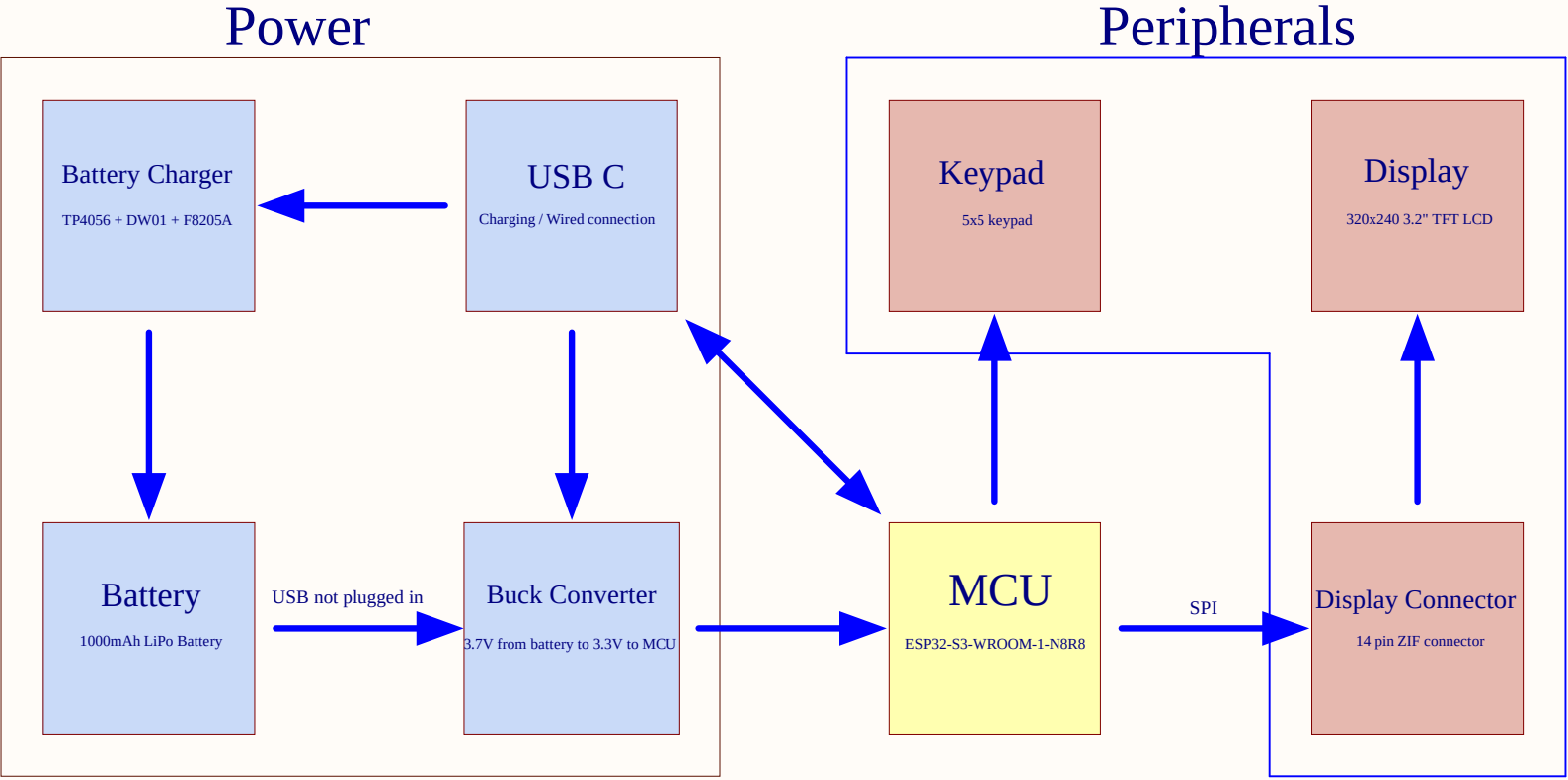
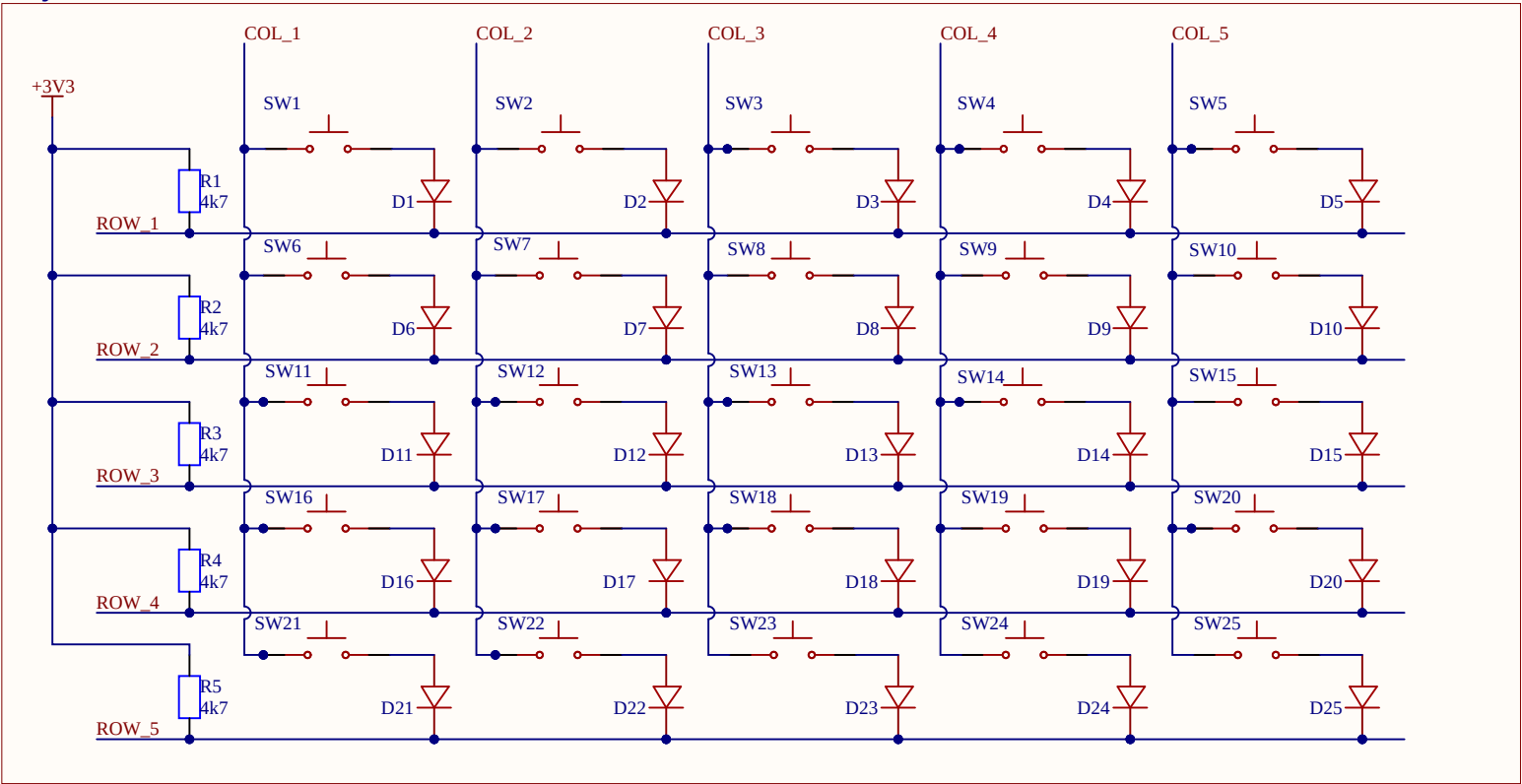


Overview

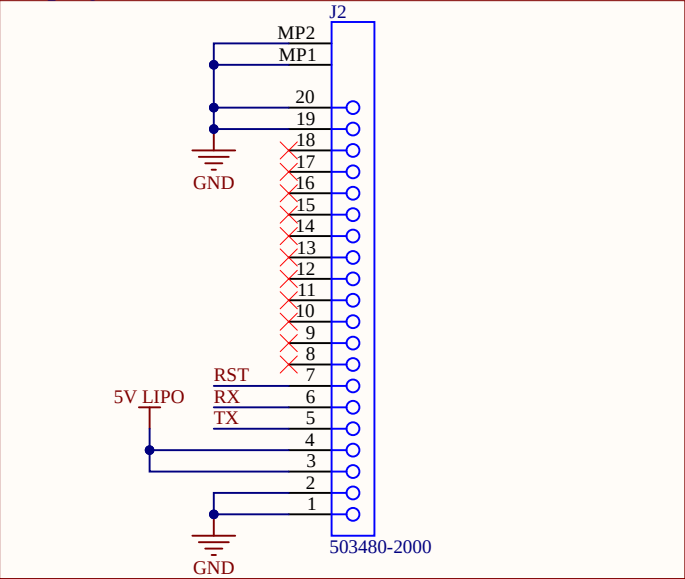


Title KEYPAD/CALCULATOR		
Size A	Number DS1	Revision 1
Date:	2/12/2026	Sheet 1 of 3
File:	C:\Users\...\OVERVIEW.SchDoc	Drawn By:

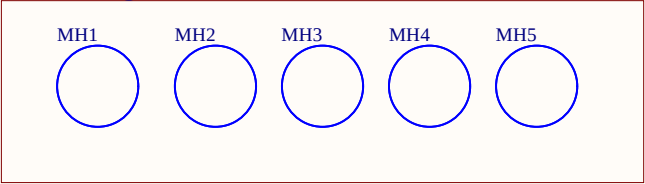
Key Matrix



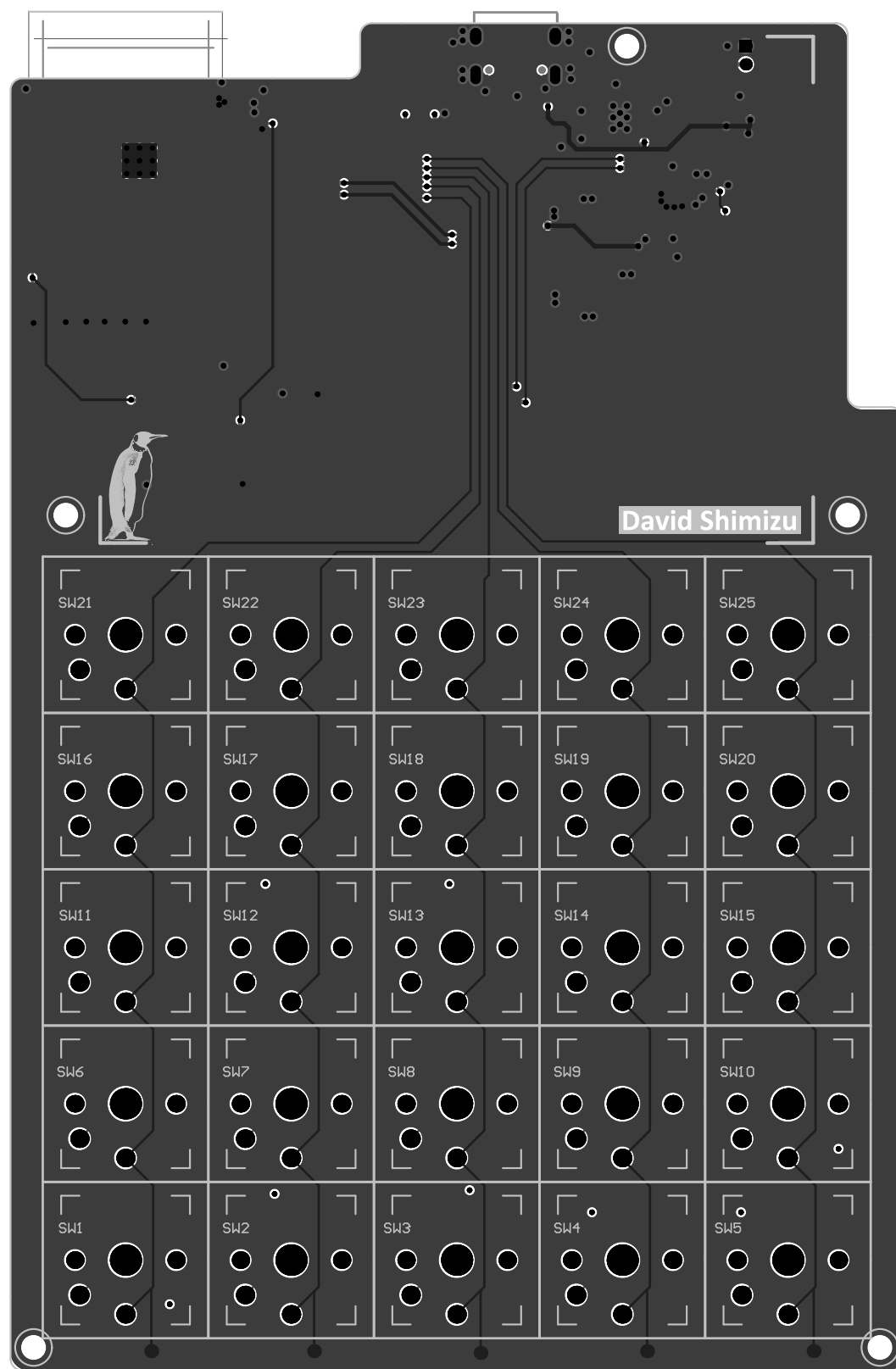
Display Connector



Mounting holes



Title		
Size	Number	Revision
A		
Date:	2/12/2026	Sheet of
File:	C:\Users\...\PERIPHERALS.SchDoc	Drawn By:



David Shimizu

SW21

SW22

SW23

SW24

SW25

SW16

SW17

SW18

SW19

SW20

SW11

SW12

SW13

SW14

SW15

SW6

SW7

SW8

SW9

SW10

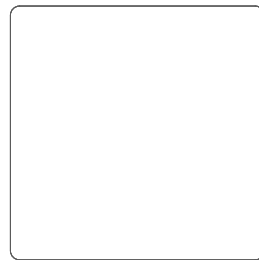
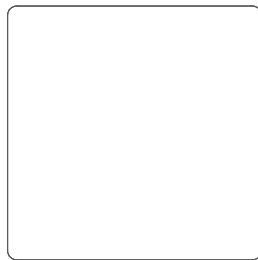
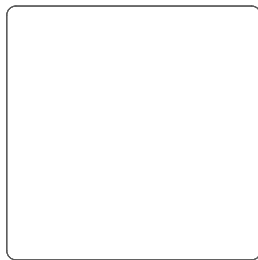
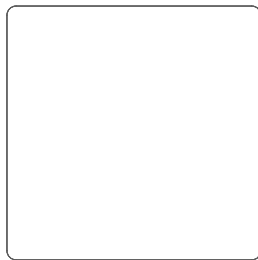
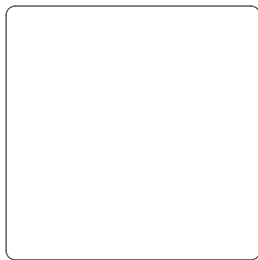
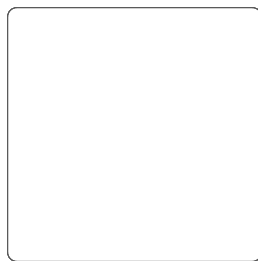
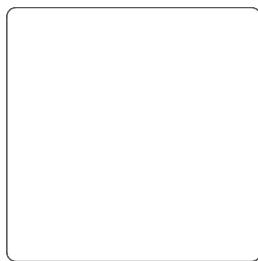
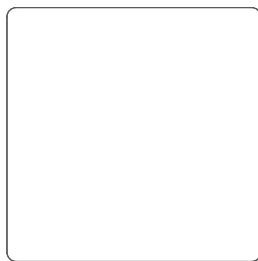
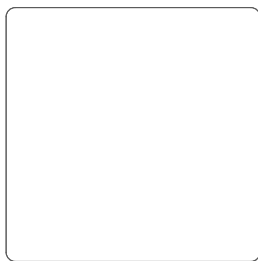
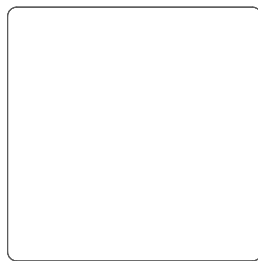
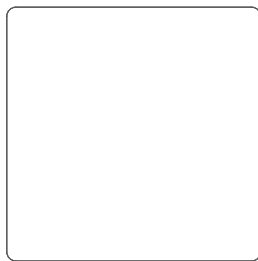
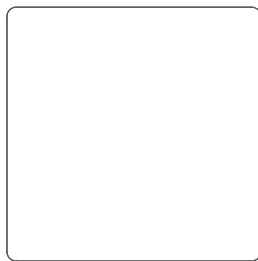
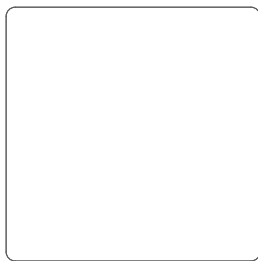
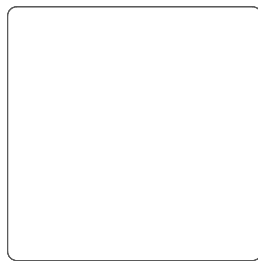
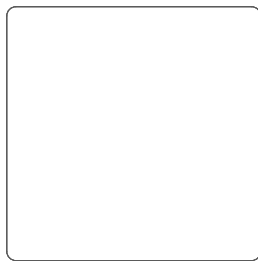
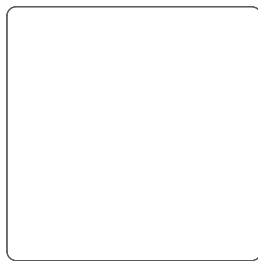
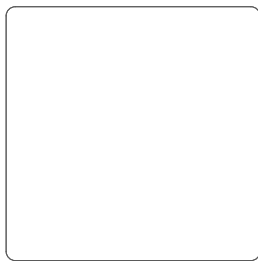
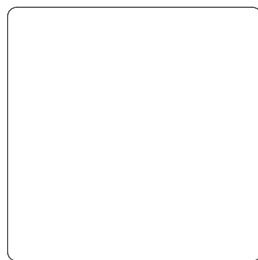
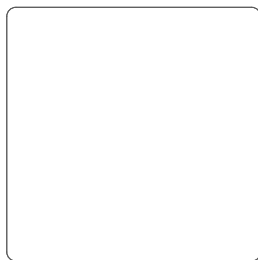
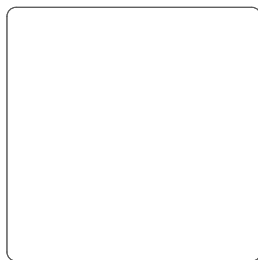
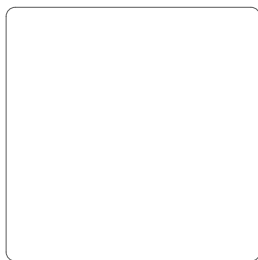
SW1

SW2

SW3

SW4

SW5



Comment	Description	Designator	Footprint	LibRef	Quantity
10uF	0402 smd capacitor	C1, C5	0402 capacitor	0402 capacitor	2
22uF	0402 smd capacitor	C2, C3	0402 capacitor	0402 capacitor	2
4u7	0402 smd capacitor	C4	0402 capacitor	0402 capacitor	1
10uF		C6, C7, C8	0603 CAPACITOR	0603 capacitor	3
100nF	0402 smd capacitor	C9, C10	0402 capacitor	0402 capacitor	2
SOD-323	1N4148WS	D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24, D25	SOD-323	SOD-323	25
RLD	Red 0603 Light Emitting Diodes	D26, D28	LED_KT-0603R	KT-0603R	2
SMF5.0AT1G	200 W Transient Voltage Suppressor, 2 Pin SOD-123FL, Pb-Free, Tape and Reel	D27	ONSC-SOD-123FL-2-498_V	OMP-1061-00612-1	1
SS14	40V Independent Type 1A 550mV@1A SMA(DO-214AQ) Schottky Diodes ROHS	D29	SMA(DO-214AQ)(schottky)	ss14	1
ESP32-S3-WROOM-1-N8R8		IC1	ESP32-S3-WROOM-1-N8R8	ESP32-S3-WROOM-1-N8R8	1
TP4056X	Integrated Circuit	IC2	SOIC127P600X1.75-9N	TP4056X	1
TPS61023DRLR	Integrated Circuit	IC3	SOTFL50P160X60-6N	TPS61023DRLR	1
battery connector		J1	2pin wire to board	2pin wire to board	1
503480-2000	0.5 mm Pitch Easy-On(TM) BackFlip(TM) Type FPC Connector, 1.0 mm Mated Height, Right Angle, SMT, Dual Contact, 20 Circuits, -40 to 85 degC, ELV and RoHS Compliant, Tape and Reel	J2	MOLX-503480-2000	OMP-1241-00121-1	1
2.2uH		L1	0603 inductor	0603 inductor	1
1uH		L2	0802 inductor	0805 inductor	1
m2 mounting hole		MH1, MH2, MH3, MH4, MH5	m2 mounting hole	m2 mounting hole	5
AO3401A	P-Channel MOSFET, 30 V, 4 A, -55 to 150 degC, 3-Pin SOT-23-3L, RoHS, Tape and Reel	Q2	AO-SOT-23-3L_V	OMP-2000-05778-1	1
4k7		R1, R2, R3, R4, R5	0603 resistor	0603 resistor	5
732K		R6	0402 resistor	0402 resistor	1
100K		R7	0402 resistor	0402 resistor	1
10K		R8, R14, R15, R16, R19	0603 resistor	0603 resistor	5
1K		R9, R17	0402 resistor	0402 resistor	2
100K		R10	0603 resistor	0603 resistor	1
5K1		R11, R12	0603 resistor	0603 resistor	2
22K		R13	0603 resistor	0603 resistor	1
1k2		R18	0603 resistor	0603 resistor	1
switch		SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8, SW9, SW10, SW11, SW12, SW13, SW14, SW15, SW16, SW17, SW18, SW19, SW20, SW21, SW22, SW23, SW24, SW25	chocv1	switch	25
push button		SW26, SW27	push button	push button	2
TLV62569DBVR	IC REG BUCK ADJ 2A SOT23-5	U1	FP-DBV0005A-MFG	OMP-04918-000387-1	1
usbc connector		USB1	USBCconnector	usbc connector	1
USBL06-2SC6		X1	STM-SOT23-6L_V	USBL06-2SC6	1